

Markscheme

Mathematics: analysis and approaches
Practice question bank

52. (a)

The value of an investment is modelled by $V = 2000e^{0.045t}$, where V is in dollars and t is the time in years.

Find the value of t when the investment first exceeds \\$5000, correct to 2 decimal places.

$$\text{set } V = 5000: 2000e^{0.045t} = 5000 \quad (M1)$$

$$e^{0.045t} = 2.5 \quad \mathbf{A1}$$

$$0.045t = \ln 2.5 \quad (M1)$$

$$t = 20.36 \quad \mathbf{A1}$$

[6 marks]

(b)

Hence determine after how many complete years the investment first exceeds \\$5000.

since $t \approx 20.36$ years, the value first exceeds \\$5000 during the 21st year (M1)

21 complete years **A1**

[2 marks]